

Transpose of matrix $\equiv$

$$\begin{matrix} n \\ m \end{matrix} \begin{matrix} A \end{matrix} = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{bmatrix}$$

$$\begin{matrix} n \\ m \end{matrix} \begin{matrix} A^T \\ \text{transpose of } A \end{matrix} = \begin{bmatrix} a_{1,1} & a_{2,1} & \dots & a_{m,1} \\ a_{1,2} & a_{2,2} & \dots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \dots & a_{m,n} \end{bmatrix}$$

$\equiv$ Full-Rank Matrix $\equiv$

(for binary matrix)

↑ the element of the matrix is $\in \{0,1\}$

• Permutation matrix $n \times n$

a square binary matrix s.t.
there is only one non-zero element
in each row and each column

e.g. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ Permutation matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow$ NOT permutation matrix

$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow$ NOT

$(A \times B^T) = B^T \times A^T$ check

full-rank matrix

a binary matrix $A \in \{0,1\}^{m \times n}$ is a full-rank matrix

if: There exists a permutation matrix P , a square matrix L and a matrix U
s.t.

$A = PLU$

$$\begin{matrix} m & n & n & m \\ P & A & = & L & U \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ & 1 & 0 & \dots & 0 \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & & 1 \end{bmatrix} \begin{bmatrix} 1 & * & \dots & * \\ & 0 & \dots & * \\ & & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \\ & & & & 0 & 0 \end{bmatrix} \Bigg\} n$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \times A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Bigg\} n=3$$

rank(A) = 3

if row(A) = n $\Rightarrow$ A is full-rank

$$m = 2n$$

$A: (a_{ij})$ rank random A is a random matrix with m rows n columns elements in $\{0,1\}$
binary matrix $\in \{0,1\}$

P_1

P_2

hopefully
($d_2 \neq d_1$)

$$\textcircled{1} f_1 \leftarrow \{0,1\}^{m \times 1}$$

$$f_1 = \begin{bmatrix} 1 \\ \vdots \\ m \end{bmatrix}$$

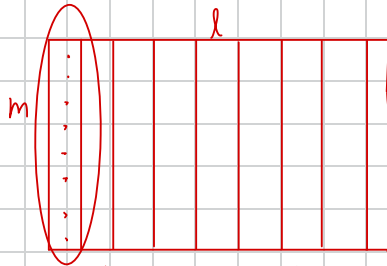
f has k non-zero element
 $n-k$ zero elements } random

$$\textcircled{2} S_1 \leftarrow \{0,1\}^{n \times l} \text{ (random)}$$

$$\textcircled{3} E_1 \leftarrow \{0,1\}^{m \times l}$$

each column has

k non-zero elements } random
 $n-k$ zero elements



k non-zero element } random
 $n-k$ zero element } for each row

create f_2, S_2, E_2 with similar procedure as f_1, S_1, E_1

$$U_2 = AS_2 + E_2 \text{ mod } 2$$

$$k = A^T f \text{ mod } 2$$

$$k_2 = U_1^T f_2 + S_2^T b_1 \text{ mod } 2$$

$$\textcircled{4} U_1 = AS_1 + E_1 \text{ mod } 2$$

$$b_1 = A^T f_1 \text{ mod } 2$$

$$\textcircled{5} k_1 = U_2^T f_1 + S_1^T b_2 \text{ mod } 2$$

Transpose

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \\ 7 & 5 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

3×2

$$A^T = \begin{bmatrix} 1 & 3 & 7 \\ -2 & 0 & 5 \end{bmatrix} \begin{matrix} 2 \times 3 \end{matrix}$$

Transpose of A

rows $\Rightarrow$ columns
columns $\Rightarrow$ row

$$A^T = \begin{bmatrix} 1 & 3 & 7 \\ -2 & 0 & 5 \end{bmatrix}$$

Properties

- ① $(A^T)^T = A$ (any)
- ② $(A+B)^T = A^T + B^T$
- ③ $(kA)^T = k(A^T)$ (Any $m \times n$)
- ④ $(AB)^T = B^T A^T$ ($m \times n$, $n \times p$)
- ⑤ $\det(A^T) = \det A$
- ⑥ $(A^{-1})^T = (A^T)^{-1}$

Proof of properties of ④

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Check: $(AB^T)_{ji}$

$$= (AB^T)_{ji}$$

$$= \sum_{k=1}^n a_{jk} b_{ki}$$

$$= \sum_{k=1}^n b_{ki} a_{jk}$$

$$= \sum_{k=1}^n (b^T)_{ik} (A^T)_{kj}$$

$$= (b^T A^T)_{ij}$$

Recall identity matrix

- Is square matrix
- Consists of only 1's and 0's
- The principal diagonal are 1's and all other elements are 0's.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrix A Matrix I Matrix A

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrix I Matrix A

$$A \times I = I \times A = A$$

Permutation Summary

Column permutation - switch columns

$$B = A \times P$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{bmatrix}$$

Matrix A Matrix P Matrix B

Row permutation - switch rows

$$B = P \times A$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

LU decomposition

$$A = \begin{pmatrix} 2 & 4 & 3 & 5 \\ -4 & -7 & -5 & -9 \\ 6 & 8 & 2 & 9 \\ 4 & 9 & -2 & 14 \end{pmatrix} \quad (4 \times 4 \text{ matrix})$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ -4 & -7 & -5 & -9 \\ 6 & 8 & 2 & 9 \\ 4 & 9 & -2 & 14 \end{pmatrix}$$

II - (-2) · I
III - 3 · I
IV - 2 · I

$$= \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & & 1 & \\ 2 & & & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & -4 & -7 & -6 \\ 0 & 1 & -8 & 4 \end{pmatrix}$$

III - (-4) · II
IV - (1) · II

$$= \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & -4 & 1 & \\ 2 & 1 & & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -9 & 2 \end{pmatrix}$$

$$L \leftarrow \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & -4 & 1 & \\ 2 & 1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & -4 \end{pmatrix} \quad \left. \vphantom{\begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & -4 & 1 & \\ 2 & 1 & 3 & 1 \end{pmatrix}} \right\} U = A$$

IV - (-3) · III

PLU decomposition
 lower triangular
 permutation matrix
 row echelon

$$A = \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix} = P_{1 \leftrightarrow 2}^2 \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ \boxed{1} & 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix}$$

$$= P_{1 \leftrightarrow 2} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ \boxed{0} & 1 & 2 & 1 & 2 \\ \boxed{2} & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix} \begin{matrix} \\ \text{II} - 0\text{I} \\ \text{III} - 2\text{I} \\ \text{IV} - 1\text{I} \end{matrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{1} & 2 & 1 & 3 \\ 0 & \boxed{2} & 4 & 3 & 5 \end{pmatrix} \begin{matrix} \\ \\ \text{III} - 1\text{II} \\ \text{IV} - 2\text{II} \end{matrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} P_{3 \leftrightarrow 4}^2 \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 1 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} P_{3 \leftrightarrow 4}^2 \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{pmatrix}$$

P L U

Rank of matrix

The rank of matrix is the number of linearly independent rows.

Or equivalently, the number of linearly independent columns. Only a zero matrix has rank zero.

To be in row echelon form, a matrix must have the following properties:

1. If a row doesn't consist entirely of zeros, then the first non-zero number in the row is a 1.
2. All zero rows are at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.

Rank matrix